

Multi-Source Candidate Data Transformer — Technical Design

Eightfold Engineering Intern (Jul–Dec 2026) Assignment · Stage 1 design, one page, no code

Pipeline

detect & parse (per-source, isolated failure, Strategy/Factory parsers) → **normalize** (email/phone-E.164/date-YYYY-MM/country-ISO3166/skill-canonical/url/name) → **merge** (deterministic conflict resolution) → **confidence** (per-field/skill/overall, computed alongside merge) → **validate** (schema sanity) → **project-to-output** (default schema or a runtime JSON config, no code changes). A bad source is isolated at parse time and never reaches the merge engine; the pipeline always returns a profile, possibly sparse, never an uncaught exception.

Canonical Schema & Normalized Formats

Exactly the assignment schema: candidate_id, full_name, emails[], phones[] (E.164), location{city, region, country (ISO3166 alpha-2)}, links{linkedin, github, portfolio, other[]}, headline, years_experience, skills[{name (canonical), confidence, sources[]}], experience[{company, title, start/end as YYYY-MM or "present", summary}], education[{institution, degree, field, end_year}], provenance[{field, source, method}], overall_confidence. Skills are canonicalized via an alias table (js→JavaScript, springboot→Spring Boot, postgres→PostgreSQL, c plus plus→C++); unknown tokens are kept (title-cased), never invented or dropped.

Sources Implemented

Structured: Recruiter CSV, ATS JSON (own field names, explicitly mapped). Unstructured: Resume (PDF via PDFBox / DOCX via POI / plain text — same regex+keyword heuristics over extracted text), Recruiter Notes (.txt). GitHub/LinkedIn are modeled in the SourceType/priority system but descoped from execution (both need a live network call, no time budget); adding either later is a pure SourceParser addition.

Merge & Conflict-Resolution Policy

Match key: one input set = one candidate. Scalar fields resolve deterministically: (1) **Consensus** — the value with the most distinct sources independently agreeing wins; (2) **Source reliability** breaks consensus ties via trust priority Recruiter CSV(1) > ATS JSON(2) > Resume(3) > GitHub(4) > LinkedIn(5) > Recruiter Notes(6); (3) **Determinism** — remaining ties broken by lexicographically smallest source id, so identical inputs always reproduce identical output. List fields (emails/phones/skills/links.other) are unions: normalized + deduplicated, every source tracked. Experience/education are deduplicated by (company,title)/(institution,degree), keeping the most complete entry (tie-break by source priority), sorted most-recent-first.

Confidence Strategy

Each source has a base trust score (CSV 0.95 ... Recruiter Notes 0.45). Field confidence = winning source's base score + a bonus per additional independently-corroborating source, capped at 0.99; a resume-only value is capped at a medium ceiling (0.70) since resumes are self-reported. Skill confidence scales with distinct corroborating sources plus a bonus if any corroborating source is structured. overall_confidence is a weighted average across whichever fields are actually present (identity + skills weighted highest), so a profile missing a low-weight field like education isn't unfairly punished.

Runtime Custom-Output Config (Projection + Validation)

CanonicalPathResolver resolves a config's "from" path (emails[0], skills[].name, location.city) against the canonical model purely by string path, so reshaping output is config-only. ProjectionEngine writes each value to the config's "path" (dot-notation nests output), re-applies a requested normalize hint (E164, canonical, ISO3166, YYYY-MM), and honors include_confidence/include_provenance toggles. on_missing is null (emit null), omit (drop key), or error (collect every missing required field, throw one exception listing all of them). No config → full canonical schema, confidence + provenance always included.

Edge Cases Handled

- Missing source file → logged + skipped, run continues with remaining sources.
- Malformed CSV / invalid JSON syntax → parser self-marks malformed, source excluded before merge.
- Same email, different casing/source → normalized + deduplicated to one canonical entry.
- Conflicting job titles across CSV vs ATS vs resume → resolved via consensus → reliability → determinism, never randomly.
- Required field missing under a strict (on_missing=error) config → one exception listing every missing field, not silent bad/partial data.

Deliberately Descoped Under Time Pressure

Live GitHub/LinkedIn fetching (needs network/auth); full NLP/NER resume parsing (regex + keyword-section heuristics used instead — weaker on heavily stylized/multi-column resumes, an explicit, documented tradeoff); a UI (assignment explicitly deprioritizes this versus a clean CLI).